



Girraween High School

Year 12 Mathematics HSC Trial Examination

2017

General Instructions

- Reading Time - 5 minutes
- Working Time - 3 hour
- Calculators and ruler may be used
- All necessary working out must be shown

Total Marks - 100

- Attempt all questions
- Marks may be deducted for careless or badly arranged work
- Start each question on a new sheet of paper

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Question 1 (1 mark)

What is 0.01029 rounded to 3 significant figures?

- A. 0.01 B. 0.010 C. 0.0103 D. 0.01029

Question 2 (1 mark)

What are the solutions to the equation $x(x - 1) = x$?

- A. $x = 0$ B. $x = 1$ C. $x = 0$ and $x = 2$ D. $x = 1$ and $x = 2$

Question 3 (1 mark)

What is the correct factorisation of $8x^3 - 1$?

- A. $(2x - 1)(4x^2 + 4x + 1)$ B. $(2x + 1)(4x^2 - 4x + 1)$ C. $(2x + 1)(4x^2 - 2x + 1)$
D. $(2x - 1)(4x^2 + 2x + 1)$

Question 4 (1 mark)

What is the angle of inclination of the line $\sqrt{3}x + y = 1$ with respect to the positive x axis?

- A. 30° B. 60° C. 120° D. 150°

Question 5 (1 mark)

Which of the following is the domain of the function $f(x) = \frac{1}{\sqrt{x+1}}$

- A. $x > -1$ B. $x \geq -1$ C. $x < -1$ D. $x \leq -1$

Question 6 (1 mark)

What is the period of the function $f(x) = 2 \sin\left(\frac{2\pi}{3}x\right)$?

- A. 3 B. 3π C. $\frac{1}{3}$ D. $\frac{\pi}{3}$

Question 7 (1 mark)

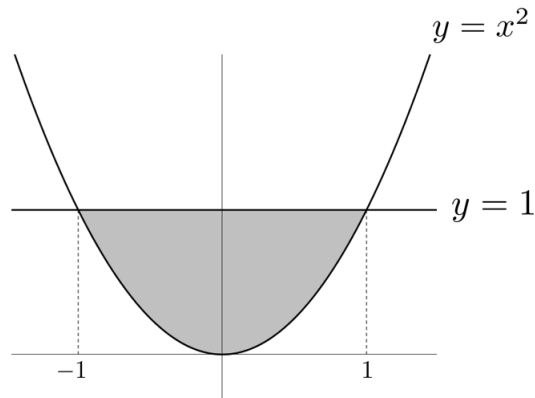
The value of the limit $\lim_{x \rightarrow 10} \frac{x^2 - 100}{x - 10}$ is

- A. undefined B. 0 C. 8 D. 20

Question 8 on the next page

Question 8 (1 mark)

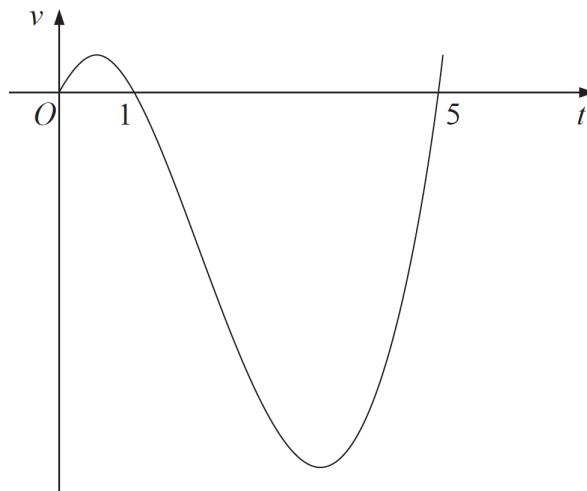
Which of the following integrals does NOT compute the area of the shaded region?



- A. $\int_{-1}^1 x^2 dx$ B. $\int_{-1}^1 1 - x^2 dx$ C. $2 - \int_{-1}^1 x^2 dx$ D. $2 \int_0^1 1 - x^2 dx$

Question 9 (1 mark)

The velocity time graph of a particle travelling in a straight line is given below. Which is the best approximate time at which the particle returns to its initial position?



- A. $t = 1$ B. $t = 2$ C. $t = 3$ D. $t = 5$

Question 10 (1 mark)

The position x of a particle travelling in a straight line is given by $x = -2t^2 + t + 1$, which of the following is false in the time frame $0 \leq t \leq 5$?

- A. The maximum speed of the particle $>$ the maximum velocity of the particle
B. The minimum speed of the particle $<$ the minimum velocity of the particle
C. The maximum speed of the particle $>$ the minimum velocity of the particle
D. The minimum speed of the particle $<$ the maximum velocity of the particle

Question 11 on the next page

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Write your answers on the paper provided.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

(a) Evaluate $\sqrt{e^2 - 1}$ to two decimal places. [1]

(b) Convert $\frac{4\pi}{3}$ radians to degrees. [1]

(c) Find the exact value of $\cos \frac{5\pi}{6}$ [1]

(d) Simplify $\frac{x}{x(x-1)} + \frac{1}{x-1}$ [2]

(e) Find the values of x for which $|x-2| \geq 5$ [2]

(f) Differentiate with respect to x

i. $y = x \ln x$ [2]

ii. $y = \frac{\sin x}{2x}$ [2]

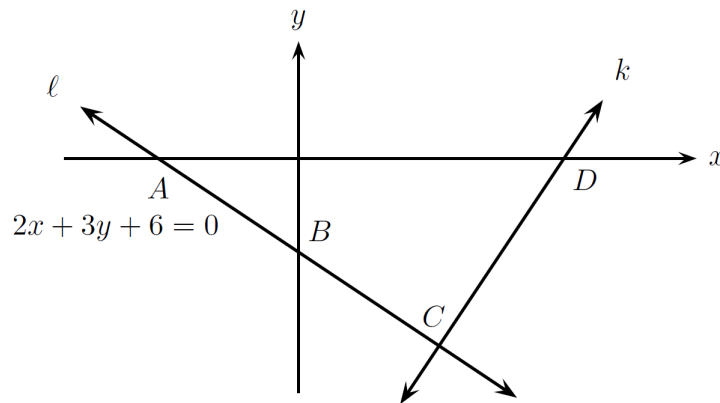
(g) Find $\int \frac{x}{x^2+1} dx$ [2]

(h) Find the value of $7 + 14 + 21 + \cdots + 175$. [2]

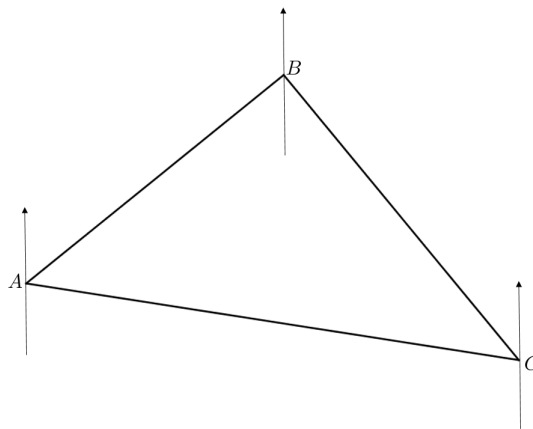
The exam continues on the next page

Question 12 (15 marks)

- (a) In the diagram below, the line ℓ has equation $2x + 3y + 6 = 0$. It cuts the x axis at A and the y axis at B and it intersects the line k at C . Line k is perpendicular to ℓ and cuts the x axis at D .



- i. Find the coordinates of A and B . [2]
 - ii. If B is the midpoint of AC show that the equation of k is given by $3x - 2y - 17 = 0$. [3]
- (b) In the diagram below, a person walks from point A on a bearing of 030° for 2.4 km to point B . From point B he walks for 3.6 km on a bearing of 145° to arrive at point C .



- i. Find the length of AC to one decimal place. [3]
 - ii. Find the bearing of point C from point A . Give your answer to the nearest minute. [2]
- (c) A school sports team has a probability of 0.2 of winning any match.
- i. Find the probability the team wins exactly one of its first two matches. [2]
 - ii. Find the least number of consecutive matches the team must play to be 90% certain that it will win at least one match. [3]

The exam continues on the next page

Question 13 (15 marks)

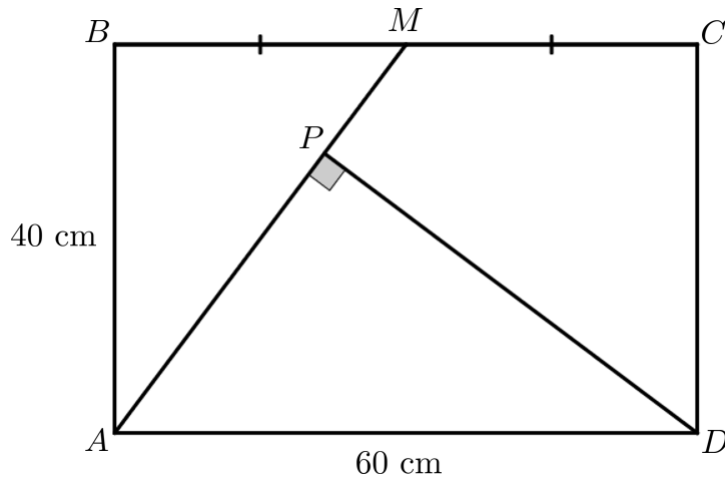
(a) Consider the parabola given by the equation $y = x^2 - 4x + 8$. Find:

i. the coordinates of the vertex. [2]

ii. the coordinates of the focus. [2]

iii. the equation of the directrix. [1]

(b) In the diagram below, $ABCD$ is a rectangle in which $AB = 40$ cm and $AD = 60$ cm. M is the midpoint of BC and DP is perpendicular to AM .



i. Prove that $\triangle ABM \parallel \triangle APD$. [3]

ii. Prove that $PD = 48$ cm. [2]

iii. Find the length of PA and hence show that the area of the quadrilateral $PMCD$ is 936 cm^2 . [2]

(c) Consider the series $2 + 4e^{-x} + 8e^{-2x} + \dots$

i. Show that the series is geometric. [1]

ii. Find the values of x such that this geometric series has an limiting sum. [1]

iii. Find the limiting sum of this geometric series in terms of x . [1]

The exam continues on the next page

Question 14 (15 marks)

(a) Consider the curve $y = x^3 - 3x + 2$.

i. Find the coordinates of the stationary points and determine their nature. [3]

ii. Find any points of inflexion. [2]

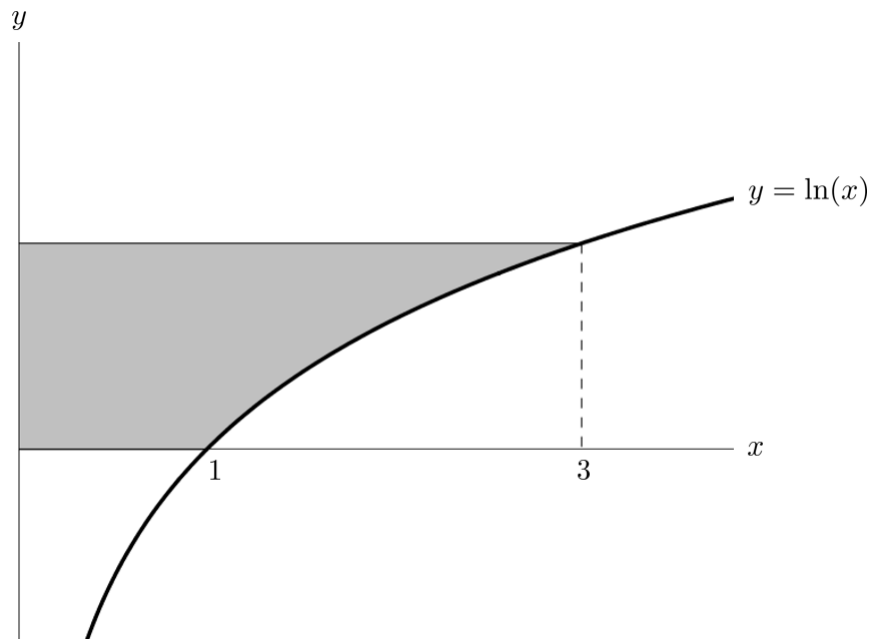
iii. Sketch the curve, showing the stationary points and any points of inflexion. [3]

iv. For what values of x is the curve increasing and concave down? [1]

(b) Use Simpson's Rule with 5 functions values to find an approximation to the value of

$$\int_1^5 \operatorname{cosec} \frac{\pi}{6} x \, dx$$

(c) Find the volume of the solid generated by rotating the shaded area below about the y -axis. [3]



The exam continues on the next page

Question 15 (15 marks)

- (a) i. Alan borrows \$10 000 at an interest rate of 12% pa, compounded monthly. [3]
Alan wants to pay off the loan with monthly repayments over 5 years. Find the required monthly repayment.

- ii. If Alan only wishes to pay \$150 each month over 5 years, how much is he able [2]
to borrow under the same interest rate?

- (b) A particle travels in a straight line. Its position x metres from the origin is given by

$$x = 6 - 2t + 8 \ln(t + 2)$$

where time t is measured in seconds.

- i. Find the initial velocity and acceleration of the particle. [2]
ii. Find the position of the particle when it is stationary. [2]
iii. State the eventual velocity and acceleration of the particle. [2]

- (c) The population N of a certain species at time t is given by

$$N = N_0 e^{-0.03t}$$

where t is in days and N_0 is the initial population of the species.

- i. Show that $N = N_0 e^{-0.03t}$ is a solution to the differential equation [1]

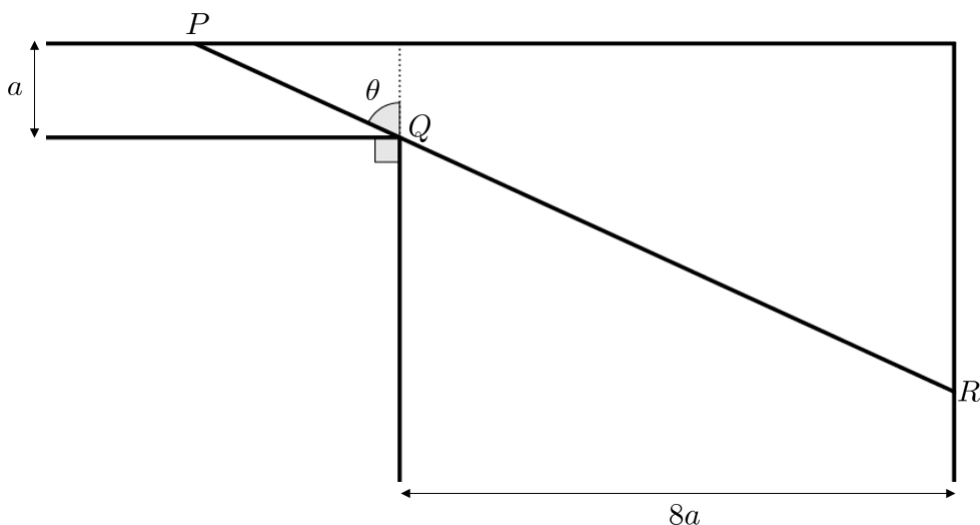
$$\frac{dN}{dt} = -0.03N$$

- ii. How long will it take for the population to halve? Give your answer to the [2]
nearest day.
iii. Find in terms of N_0 , the rate of change of the population when the population [1]
has halved.

The exam continues on the next page

Question 16 (15 marks)

- (a) Two corridors of widths a and $8a$ intersect at a right angle. A straight rope PR touches the corner Q . Let L be the length of PR .



- i. Show that $L = \frac{a}{\cos \theta} + \frac{8a}{\sin \theta}$ [1]
 - ii. Show that $\frac{dL}{d\theta} = \frac{a \cos \theta}{\sin^2 \theta} (\tan^3 \theta - 8)$ [2]
 - iii. Find the exact value of the minimum value of L . [3]
- (b) Suppose α and β are roots of $ax^2 + bx + c = 0$ where $a \neq 0$ and $\alpha \geq \beta$.
- i. Show that $\alpha - \beta = \frac{\sqrt{\Delta}}{a}$ where $\Delta = b^2 - 4ac$. [1]
 - ii. Show that $\alpha^2 - \beta^2 = -\frac{b\sqrt{\Delta}}{a^2}$. [1]
 - iii. Show that $\alpha^3 - \beta^3 = \frac{\sqrt{\Delta}(b^2 - ac)}{a^3}$. [2]
 - iv. Show that $\int_{\beta}^{\alpha} ax^2 + bx + c \, dx = -\frac{\Delta^{3/2}}{6a^2}$ [3]
- (c) Find the values of m such that equation $e^x = mx$ has no solutions. [2]

End of exam

2017 2H TRIALS

C C D C A

A D A B B

Q1

$$= 0.0103 \therefore \textcircled{C}$$

Q2

$$x(x-1) = x$$

$$x^2 - x = x$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$\therefore x = 0 \text{ \& } x = 2 \therefore \textcircled{C}$$

Q3

$$8t^3 - 1 = (2t)^3 - 1$$

$$= (2t-1)(4t^2 + 2t + 1) \therefore \textcircled{D}$$

Q4

$$g = -\sqrt{3}n + 1$$

$$\therefore \tan \theta = -\sqrt{3} \therefore \theta = 120^\circ \therefore \textcircled{C}$$

Q5

$$x+1 > 0 \therefore x > -1 \therefore \textcircled{A}$$

Q6

$$T = \frac{2\pi}{\frac{2\pi}{3}} = 3 \therefore \textcircled{A}$$

Q7

$$\lim_{x \rightarrow 10} \frac{(x+10)(x-10)}{x-10}$$

$$= \lim_{x \rightarrow 10} x+10 = 20 \therefore \textcircled{D}$$

Q8

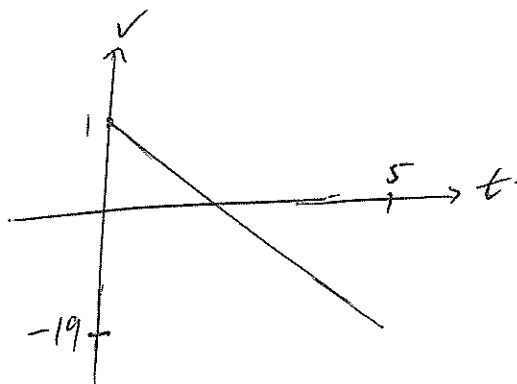
$\textcircled{A}$

Q9

$$t = 2 \textcircled{B}$$

Q10

$$V = -4t + 1$$



$$\text{Max } V = 1 \quad \text{Max } S = 19$$

$$\text{Min } V = -19 \quad \text{Min } S = 0$$

$\therefore \textcircled{B}$

Q11

$$(a) 2.53 \text{ (2 dp)}$$

$$(b) \frac{4 \times 180}{3} = 240^\circ$$

$$(c) \cos\left(\frac{5\pi}{6}\right) = -\cos\frac{\pi}{6} = -\frac{\sqrt{3}}{2}$$

$$(d) = \frac{x}{x(x-1)} + \frac{x}{x(x-1)} = \frac{2}{x-1}$$

$$(e) \begin{aligned} x-2 &\geq 5 \text{ and } 2-x \geq 5 \\ x &\geq 7 \text{ and } x \leq -3 \end{aligned}$$

Q11

(f)

$$(i) y = x \ln x$$

$$u = x \quad v = \ln x$$

$$u' = 1 \quad v' = \frac{1}{x}$$

$$y' = \ln x + 1$$

$$(ii) y = \frac{\sin x}{2x}$$

$$u = \sin x \quad v = 2x$$

$$u' = \cos x \quad v' = 2$$

$$y' = \frac{2x \cos x - 2 \sin x}{4x^2}$$

$$(g) \int \frac{x}{x^2+1} dx$$

$$= \frac{1}{2} \int \frac{2x}{x^2+1} dx = \frac{1}{2} \ln(x^2+1) + C.$$

$$(h) 7 + 14 + \dots + 175$$

$$175 = 25 \times 7$$

$$\therefore a = 7 \quad n = 25$$

$$S_{25} = \frac{25}{2} (7 + 175) = 2275$$

Q12

$$(a) (i) 2x + 3y + 6 = 0$$

$$y = 0, x = -3 \therefore A = (-3, 0)$$

$$x = 0, y = -2 \therefore B = (0, -2).$$

(a)

$$(ii) \text{ Let } C = (x, y)$$

$$\therefore \frac{-3+x}{2} = 0 \quad \& \quad \frac{0+y}{2} = -2$$

$$\therefore x = 3. \quad \therefore y = -4.$$

$$\therefore C = (3, -4).$$

$$3y = -2x - 6$$

$$y = -\frac{2}{3}x - 2$$

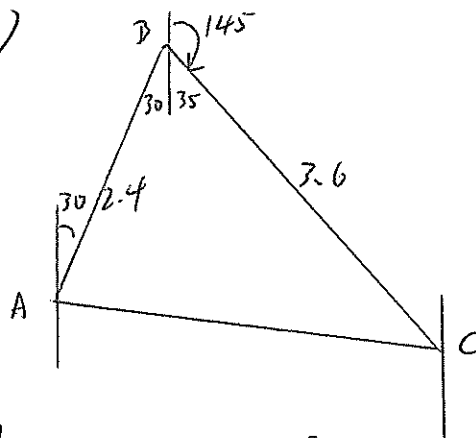
$$\therefore \text{ m of } k \text{ is } \frac{3}{2}$$

$$\therefore y + 4 = \frac{3}{2}(x - 3)$$

$$2y + 8 = 3x - 9.$$

$$\therefore 3x - 2y - 17 = 0.$$

(b)



$$(i) AC^2 = 2.4^2 + 3.6^2 - 2 \times 2.4 \times 3.6 \times \cos 65^\circ$$

$$AC = 3.8 \text{ km (labeled as 3.8)}$$

$$(ii) \frac{\sin(\angle BAC)}{3.6} = \frac{\sin 65^\circ}{AC}$$

$$\angle BAC = \sin^{-1} \left(\frac{3.6 \sin 65^\circ}{AC} \right) = 74^\circ 56'$$

$$\therefore \text{ Bearing} = 104^\circ 56'$$

Q12

$$(c) P(WL) + P(LW)$$

$$(i) = 0.2 \times 0.8 + 0.8 \times 0.2$$

$$= 0.32$$

(ii) Let n be the number of matches needed.

$$\therefore P(\text{Win at least 1}) = 1 - P(\text{all lose})$$

$$= 1 - 0.8^n$$

$$\text{We need } 1 - 0.8^n = 0.9$$

$$\therefore 0.8^n = 0.1$$

$$n = \log_{0.8}(0.1)$$

$$n = \frac{\ln 0.1}{\ln 0.8} = 10.31 \dots$$

$\therefore$ need at least 11 games.

Q13

(a)

$$(i) y - 8 = x^2 - 4x$$

$$y - 4 = x^2 - 4x + 4$$

$$y - 4 = (x - 2)^2$$

$$(x - 2)^2 = 4\left(\frac{1}{4}\right)(y - 4)$$

$$\therefore V = (2, 4)$$

(ii)



$$\therefore F = (2, 4\frac{1}{4})$$

$$(iii) g = 3\frac{3}{4}$$

(b)

$$(i) \angle ABM = \angle APD = 90$$

($\angle$ in a rectangle)

$$\angle BMA = \angle PAD$$

(Alternate $\angle$'s, $BC \parallel AD$)

$$\therefore \triangle ABM \cong \triangle APD \text{ (Equirangular)}$$

$$(ii) BM = 30 \therefore AM = 50$$

$$\frac{PD}{AB} = \frac{AD}{AM} \text{ (ratio of matching sides of } \cong \Delta \text{'s)}$$

$$\therefore \frac{PD}{40} = \frac{60}{50}$$

$$\therefore PD = 48 \text{ cm}$$

$$(iii) PA^2 = 60^2 - 48^2$$

$$\therefore PA = 36$$

$$\text{Area of } PMCD = 40 \times 60 - \text{Area of } 2 \Delta \text{'s}$$

$$= 40 \times 60 - \frac{1}{2}(40 \times 30 + 48 \times 36)$$

$$= 936$$

Q13

(c)

$$(i) \frac{T_2}{T_1} = \frac{4e^{-x}}{2} = 2e^{-x}$$

$$\frac{T_3}{T_2} = \frac{8e^{-2x}}{4e^{-x}} = 2e^{-x}$$

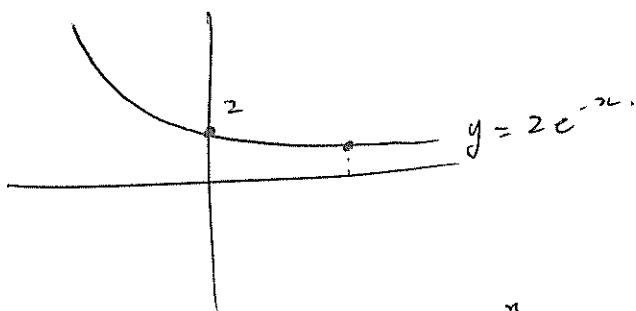
$$\therefore r = 2e^{-x} \therefore \text{is a GP.}$$

(ii) Need $-1 < r < 1$

$$\therefore -1 < 2e^{-x} < 1$$

But since $e^{-x} > 0$ for all x

$$\therefore -1 < 2e^{-x} \text{ is true for all } x$$



we check where $2e^{-x} = 1$

$$\therefore e^{-x} = \frac{1}{2}$$

$$-x = \ln \frac{1}{2}$$

$$\therefore x = \ln 2$$

$$\therefore 2e^{-x} < 1 \text{ when } x > \ln 2$$

$\therefore$ for $x > \ln 2$ limiting sum exists.

(iii)

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{2}{1-2e^{-x}}$$

$$S_{\infty} = \frac{2e^x}{e^x - 2}$$

Q14

(k)

$$(i) y = x^3 - 3x + 2$$

$$y' = 3x^2 - 3 = 3(x-1)(x+1)$$

$$y'' = 6x$$

$$y' = 0 \text{ when } x = \pm 1$$

$$y(1) = 0 \quad y(-1) = 4$$

$\therefore$ Stationary pts are $(1, 0)$ & $(-1, 4)$

$$y''(1) = 6 > 0 \therefore (1, 0) \text{ is min}$$

$$y''(-1) = -6 < 0 \therefore (-1, 4) \text{ is max.}$$

$$(ii) y'' = 0 \text{ when } x = 0$$

$$y(0) = 2 \therefore (0, 2) \text{ is possible point of inflexion.}$$

x	-1	0	1
y''	-6	0	6

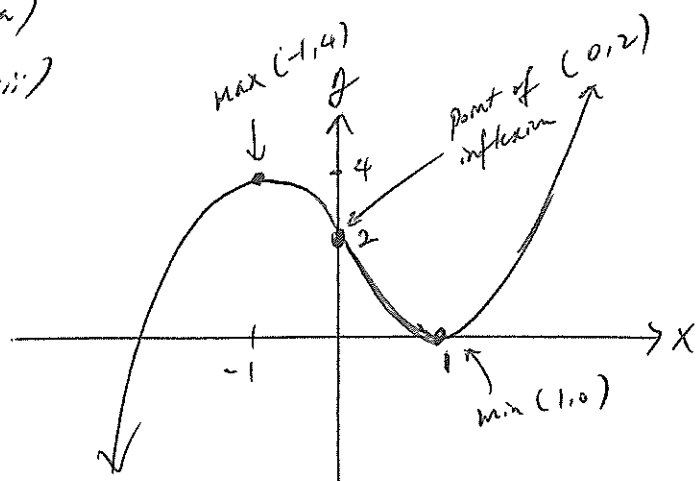
$\therefore$ concavity changes

$\therefore (0, 2)$ is a point of inflexion.

Q14

(a)

(iii)

(iv) $x < -1$

$$(b) h = \frac{5-1}{4} = 1.$$

$$I \approx \frac{1}{3} [f(1) + 4f(2) + 2f(3) + 4f(4) + f(5)]$$

$$\approx \frac{1}{3} \left[2 + 4\left(\frac{2}{\sqrt{3}}\right) + 2(1) + 4\left(\frac{2}{\sqrt{3}}\right) + 2 \right]$$

$$\approx 5.08 \text{ (2dp).}$$

$$(c) y = \ln x$$

$$\therefore x = e^y$$

$$x^2 = e^{2y}$$

$$V = \pi \int_0^{\ln 3} e^{2y} dy$$

$$V = \frac{\pi}{2} \left[e^{2y} \right]_0^{\ln 3}$$

$$V = \frac{\pi}{2} [e^{2\ln 3} - 1]$$

$$V = \frac{\pi}{2} [9 - 1] = 4\pi.$$

Q15

(a)

i) Let A_n denote amount
owe by end of n^{th} month.

$$r = \frac{12}{12} \div 100 = 0.01$$

$$A_1 = 10000(1.01) - M$$

$$A_2 = 10000(1.01)^2 - M(1.01) - M$$

$$A_3 = 10000(1.01)^3 - M(1.01)^2 - M(1.01) - M$$

$$A_3 = 10000(1.01)^3 - M(1 + 1.01 + 1.01^2)$$

$$\therefore A_n = 10000(1.01)^n - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{n-1})$$

$$\therefore A_{60} = 10000(1.01)^{60} - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{59})$$

$$A_{60} = 0$$

$$\therefore 10000(1.01)^{60} - M \left[\frac{1.01^{60} - 1}{0.01} \right] = 0$$

$$\therefore M = \frac{10000(1.01)^{60} \times 0.01}{1.01^{60} - 1}$$

$$M = \$222.44$$

Q15

(a)

$$(i) P(1.01)^{60} - 150 \left[\frac{1.01^{60} - 1}{0.01} \right] = 0.$$

$$\therefore P = \frac{150(1.01^{60} - 1)}{0.01 \times (1.01)^{60}}$$

$$P = \$6743.26$$

(b)

$$(i) x = 6 - 2t + 8 \ln(t+2)$$

$$v = -2 + \frac{8}{t+2} = -2 + 8(t+2)^{-1}$$

$$a = -8(t+2)^{-2} = -\frac{8}{(t+2)^2}$$

$$v(0) = 2 \text{ ms}^{-1}$$

$$a(0) = -2 \text{ ms}^{-2}$$

$$(ii) v=0 \text{ when } -2 + \frac{8}{t+2} = 0.$$

$$\therefore t+2 = 4 \therefore t = 2.$$

$$x(2) = 6 - 4 + 8 \ln 4$$

$$x = 2 + 8 \ln 4 \text{ m}$$

$$(iv) \text{ as } t \rightarrow \infty \quad v \rightarrow -2 \text{ ms}^{-1}$$

$$a \rightarrow 0 \text{ ms}^{-2}$$

Q15

(c)

$$(i) N = N_0 e^{-0.03t}.$$

$$\frac{dN}{dt} = -0.03 \times N_0 e^{-0.03t}.$$

$$\therefore \frac{dN}{dt} = -0.03 N$$

(ii)

$$\frac{N_0}{2} = N_0 e^{-0.03t}.$$

$$\therefore e^{-0.03t} = \frac{1}{2}.$$

$$-0.03t = \ln \frac{1}{2}$$

$$0.03t = \ln 2$$

$$t = \frac{\ln 2}{0.03} = 23.104 \dots$$

$$\therefore 24 \text{ days needed.}$$

(iii)

$$\frac{dN}{dt} = -0.03 N$$

$$\text{So when } N = \frac{N_0}{2} \quad \frac{dN}{dt} = -0.03 \times \frac{N_0}{2}$$

$$\frac{dN}{dt} = \frac{3}{200} N_0$$

(iv)

Q16

(a)

$$(i) \cos \theta = \frac{a}{pQ} \quad \sin \theta = \frac{8a}{QR}$$

$$\therefore pQ = \frac{a}{\cos \theta} \quad \therefore QR = \frac{8a}{\sin \theta}$$

$$L = pQ + QR = \frac{a}{\cos \theta} + \frac{8a}{\sin \theta}$$

$$(ii) L = a(\cos \theta)^{-1} + 8a(\sin \theta)^{-1}$$

$$\frac{dL}{d\theta} = -a(\cos \theta)^{-2} \times -\sin \theta - 8a(\sin \theta)^{-2} \times \cos \theta$$

$$\frac{dL}{d\theta} = \frac{a \sin \theta}{\cos^2 \theta} - \frac{8a \cos \theta}{\sin^2 \theta}$$

$$\frac{dL}{d\theta} = \frac{a \cos \theta}{\sin^2 \theta} \left(\frac{\sin^3 \theta}{\cos^3 \theta} - 8 \right)$$

$$\frac{dL}{d\theta} = \frac{a \cos \theta}{\sin^2 \theta} (\tan^3 \theta - 8)$$

$$(iii) \frac{dL}{d\theta} = 0 \text{ when } \tan^3 \theta - 8 = 0$$

$$\therefore \tan \theta = 2$$

note that $\cos \theta \neq 0$ for $0 < \theta < \frac{\pi}{2}$

$\therefore \theta = \tan^{-1} 2$ gives the stationary point.

Now as $\tan \theta$ is an increasing function

if $\theta > \tan^{-1} 2$ then $\tan \theta > 2$

$$\therefore \tan^3 \theta > 8$$

$$\therefore \frac{dL}{d\theta} > 0$$

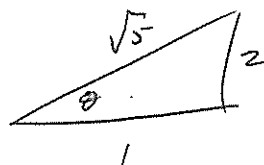
if $\theta < \tan^{-1} 2$ then $\tan \theta < 2$

$$\therefore \tan^3 \theta < 8$$

$$\therefore \frac{dL}{d\theta} < 0$$

$\therefore \theta = \tan^{-1} 2$ gives min L .

Now if $\tan \theta = 2$



$$\therefore \sin \theta = \frac{2}{\sqrt{5}} \quad \& \quad \cos \theta = \frac{1}{\sqrt{5}}$$

$$L = \frac{a}{\cos \theta} + \frac{8a}{\sin \theta}$$

$$\therefore L = \sqrt{5}a + 8a\left(\frac{\sqrt{5}}{2}\right)$$

$$L = \sqrt{5}a + 4a\sqrt{5}$$

$$L = 5a\sqrt{5}$$

(b)

$$(i) \alpha = \frac{-b + \sqrt{\Delta}}{2a} \quad \beta = \frac{-b - \sqrt{\Delta}}{2a}$$

$$\begin{aligned} \therefore \alpha - \beta &= \frac{-b + \sqrt{\Delta} + b + \sqrt{\Delta}}{2a} \\ &= \frac{\sqrt{\Delta}}{a} \end{aligned}$$

$$\begin{aligned} (ii) \alpha^2 - \beta^2 &= (\alpha - \beta)(\alpha + \beta) \\ &= \frac{\sqrt{\Delta}}{a} \times \frac{-b}{a} = -\frac{b\sqrt{\Delta}}{a^2} \end{aligned}$$

Q16

(b)

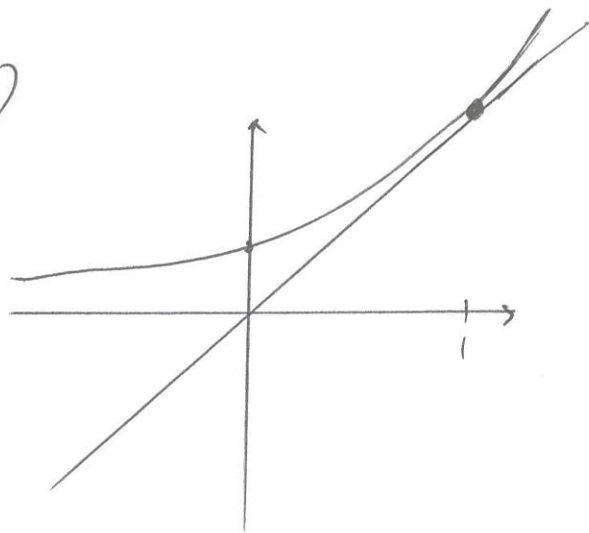
(iii)

$$\begin{aligned} \alpha^3 - \beta^3 &= (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2) \\ &= (\alpha - \beta)(\alpha^2 + 2\alpha\beta + \beta^2 - \alpha\beta) \\ &= (\alpha - \beta)[(\alpha + \beta)^2 - \alpha\beta] \\ &= \frac{\sqrt{\Delta}}{a} \left[\frac{b^2}{a^2} - \frac{c}{a} \right] \\ &= \frac{\sqrt{\Delta}}{a} \left(\frac{b^2 - ac}{a^2} \right) = \frac{\sqrt{\Delta}(b^2 - ac)}{a^3} \end{aligned}$$

(iv)

$$\begin{aligned} &\int_{\beta}^{\alpha} (an^2 + bn + c) dn \\ &= \left[\frac{an^3}{3} + \frac{bn^2}{2} + cn \right]_{\beta}^{\alpha} \\ &= \frac{a\alpha^3}{3} + \frac{b\alpha^2}{2} + c\alpha - \frac{a\beta^3}{3} - \frac{b\beta^2}{2} - c\beta \\ &= \frac{a}{3}(\alpha^3 - \beta^3) + \frac{b}{2}(\alpha^2 - \beta^2) + c(\alpha - \beta) \\ &= \frac{a}{3} \times \frac{\sqrt{\Delta}(b^2 - ac)}{a^3} - \frac{b}{2} \times \frac{b\sqrt{\Delta}}{a^2} + c \left(\frac{\sqrt{\Delta}}{a} \right) \\ &= \frac{\sqrt{\Delta}(b^2 - ac)}{3a^2} - \frac{b^2\sqrt{\Delta}}{2a^2} + \frac{c\sqrt{\Delta}}{a} \\ &= \frac{2\sqrt{\Delta}b^2 - 2ac\sqrt{\Delta} - 3b^2\sqrt{\Delta} + 6ac\sqrt{\Delta}}{6a^2} \\ &= \frac{4ac\sqrt{\Delta} - b^2\sqrt{\Delta}}{6a^2} = \frac{\sqrt{\Delta}(4ac - b^2)}{6a^2} = \frac{-\Delta^{3/2}}{6ac} \end{aligned}$$

(c)



We first find the value of m such that $y = mx$ is tangent to $y = e^x$

$$\therefore m = e^x \dots (1)$$

$$mx = e^x \dots (2)$$

$$\frac{(2)}{(1)} \text{ gives } x = 1$$

$$\therefore m = e.$$

$\therefore$ no solutions i.e. no intersections when

$$0 \leq m < e$$